

## SAFETY DATA SHEET

Creation Date 23-December-2009

Revision Date 29-March-2024

Revision Number 3

### 1. Identification

**Product Name** Nickel(II) nitrate hexahydrate

**Cat No. :** 10816

**CAS-No** 13478-00-7  
**Synonyms** Nickelous nitrate hexahydrate

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

##### **Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

##### **WHMIS 2015 Classification**

Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                             |             |
|-------------------------------------------------------------|-------------|
| <b>Oxidizing solids</b>                                     | Category 2  |
| <b>Acute oral toxicity</b>                                  | Category 4  |
| <b>Acute Inhalation Toxicity</b>                            | Category 4  |
| <b>Skin Corrosion/Irritation</b>                            | Category 2  |
| <b>Serious Eye Damage/Eye Irritation</b>                    | Category 1  |
| <b>Respiratory Sensitization</b>                            | Category 1  |
| <b>Skin Sensitization</b>                                   | Category 1  |
| <b>Germ Cell Mutagenicity</b>                               | Category 2  |
| <b>Carcinogenicity</b>                                      | Category 1A |
| <b>Reproductive Toxicity</b>                                | Category 1B |
| <b>Specific target organ toxicity (single exposure)</b>     | Category 3  |
| Target Organs - Respiratory system.                         |             |
| <b>Specific target organ toxicity - (repeated exposure)</b> | Category 1  |
| Target Organs - Lungs.                                      |             |

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**Label Elements****Signal Word**

Danger

**Hazard Statements**

May intensify fire; oxidizer  
Harmful if swallowed or if inhaled  
Causes skin irritation  
May cause an allergic skin reaction  
Causes serious eye damage  
May cause allergy or asthma symptoms or breathing difficulties if inhaled  
May cause respiratory irritation  
Suspected of causing genetic defects  
May cause cancer by inhalation  
May damage the unborn child  
Causes damage to organs through prolonged or repeated exposure  
Harmful if inhaled

**Precautionary Statements****Prevention**

Obtain special instructions before use  
Do not handle until all safety precautions have been read and understood  
Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
Keep/Store away from clothing/combustible materials  
Take any precaution to avoid mixing with combustibles  
Do not breathe dust/fumes/gas/mist/vapours/spray  
Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Use only outdoors or in a well-ventilated area  
Contaminated work clothing should not be allowed out of the workplace  
Wear protective gloves/protective clothing/eye protection/face protection  
Wear respiratory protection

**Response**

IF ON SKIN: Wash with plenty of soap and water  
IF INHALED: Remove person to fresh air and keep comfortable for breathing  
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
Immediately call a POISON CENTER/doctor  
Rinse mouth  
If experiencing respiratory symptoms: Call a POISON CENTER/doctor  
In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish  
Take off contaminated clothing and wash it before reuse

**Storage**

Store locked up  
Store in a well-ventilated place. Keep container tightly closed

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Other Hazards**

Very toxic to aquatic life with long lasting effects

### 3. Composition/Information on Ingredients

| Component                               | CAS-No     | Weight % |
|-----------------------------------------|------------|----------|
| Nickel(II) nitrate, hexahydrate (1:2:6) | 13478-00-7 | >95      |
| Nickel(II) nitrate                      | 13138-45-9 | -        |

### 4. First-aid measures

|                                        |                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                     | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                       |
| <b>Skin Contact</b>                    | Wash off immediately with plenty of water for at least 15 minutes. Get medical attention.                                                                                                                                                                                                                               |
| <b>Inhalation</b>                      | Remove to fresh air. If breathing is difficult, give oxygen. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required. |
| <b>Ingestion</b>                       | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                          |
| <b>Most important symptoms/effects</b> | Causes eye burns. May cause allergy or asthma symptoms or breathing difficulties if inhaled. May cause allergic skin reaction. . Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing  |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                                                                                                                                                                                                                                   |

### 5. Fire-fighting measures

|                                         |                                                                |
|-----------------------------------------|----------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Use extinguishing agent suitable for type of surrounding fire. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                       |
| <b>Flash Point</b>                      | No information available                                       |
| <b>Method -</b>                         | No information available                                       |
| <b>Autoignition Temperature</b>         | No information available                                       |
| <b>Explosion Limits</b>                 |                                                                |
| <b>Upper</b>                            | No data available                                              |
| <b>Lower</b>                            | No data available                                              |
| <b>Oxidizing Properties</b>             | Oxidizer                                                       |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                       |
| <b>Sensitivity to Static Discharge</b>  | No information available                                       |

#### Specific Hazards Arising from the Chemical

Oxidizer: Contact with combustible/organic material may cause fire. Thermal decomposition can lead to release of irritating gases and vapors. Keep product and empty container away from heat and sources of ignition. May ignite combustibles (wood paper, oil, clothing, etc.). Do not allow run-off from fire-fighting to enter drains or water courses.

#### Hazardous Combustion Products

nitric acid.

#### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

#### NFPA

Health  
3Flammability  
1Instability  
2Physical hazards  
OX

## 6. Accidental release measures

### Personal Precautions

Use personal protective equipment as required. Evacuate personnel to safe areas. Ensure adequate ventilation. Avoid dust formation. Avoid contact with skin, eyes or clothing.

### Environmental Precautions

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained. See Section 12 for additional Ecological Information. Avoid release to the environment. Collect spillage.

### Methods for Containment and Clean Up

Keep combustibles (wood, paper, oil, etc) away from spilled material. Sweep up and shovel into suitable containers for disposal. Avoid dust formation. Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Sweep up and shovel into suitable containers for disposal.

## 7. Handling and storage

### Handling

Use only under a chemical fume hood. Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Keep away from clothing and other combustible materials. Avoid dust formation. Do not breathe (dust, vapor, mist, gas). Do not ingest. If swallowed then seek immediate medical assistance.

### Storage.

Keep containers tightly closed in a dry, cool and well-ventilated place. Do not store near combustible materials. Incompatible Materials. Strong oxidizing agents. Organic materials. Finely powdered metals. Acids. Strong reducing agents. Combustible material.

## 8. Exposure controls / personal protection

### Exposure Guidelines

| Component                               | Alberta                    | British Columbia            | Ontario TWAEV              | Quebec                     | ACGIH TLV                  | OSHA PEL                             | NIOSH                                                      |
|-----------------------------------------|----------------------------|-----------------------------|----------------------------|----------------------------|----------------------------|--------------------------------------|------------------------------------------------------------|
| Nickel(II) nitrate, hexahydrate (1:2:6) | TWA: 0.1 mg/m <sup>3</sup> | TWA: 0.05 mg/m <sup>3</sup> | TWA: 0.1 mg/m <sup>3</sup> | TWA: 0.1 mg/m <sup>3</sup> | TWA: 0.1 mg/m <sup>3</sup> | (Vacated) TWA: 0.1 mg/m <sup>3</sup> | IDLH: 10 mg/m <sup>3</sup><br>TWA: 0.015 mg/m <sup>3</sup> |
| Nickel(II) nitrate                      | TWA: 0.1 mg/m <sup>3</sup> | TWA: 0.05 mg/m <sup>3</sup> | TWA: 0.1 mg/m <sup>3</sup> | TWA: 0.1 mg/m <sup>3</sup> | TWA: 0.1 mg/m <sup>3</sup> | (Vacated) TWA: 0.1 mg/m <sup>3</sup> | IDLH: 10 mg/m <sup>3</sup><br>TWA: 0.015 mg/m <sup>3</sup> |

#### Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

### Engineering Measures

Use only under a chemical fume hood. Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles

#### Hand Protection

Wear appropriate protective gloves and clothing to prevent skin exposure.

| Glove material | Breakthrough time                 | Glove thickness | Glove comments         |
|----------------|-----------------------------------|-----------------|------------------------|
| Natural rubber | See manufacturers recommendations |                 | Splash protection only |
| Nitrile rubber |                                   |                 |                        |

Neoprene  
PVC

-

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Particulates filter conforming to EN 143

When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls**

Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 9. Physical and chemical properties

|                                               |                          |
|-----------------------------------------------|--------------------------|
| <b>Physical State</b>                         | Solid                    |
| <b>Appearance</b>                             | Blue green               |
| <b>Odor</b>                                   | Odorless                 |
| <b>Odor Threshold</b>                         | No information available |
| <b>pH</b>                                     | 5 50g/L (20°C)           |
| <b>Melting Point/Range</b>                    | 56.7 °C / 134.1 °F       |
| <b>Boiling Point/Range</b>                    | 137 °C / 278.6 °F        |
| <b>Flash Point</b>                            | No information available |
| <b>Evaporation Rate</b>                       | Not applicable           |
| <b>Flammability (solid,gas)</b>               | No information available |
| <b>Flammability or explosive limits</b>       |                          |
| <b>Upper</b>                                  | No data available        |
| <b>Lower</b>                                  | No data available        |
| <b>Vapor Pressure</b>                         | negligible               |
| <b>Vapor Density</b>                          | Not applicable           |
| <b>Specific Gravity</b>                       | No information available |
| <b>Solubility</b>                             | Soluble in water         |
| <b>Partition coefficient; n-octanol/water</b> | No data available        |
| <b>Autoignition Temperature</b>               | No information available |
| <b>Decomposition Temperature</b>              | 137 °C                   |
| <b>Viscosity</b>                              | Not applicable           |
| <b>Molecular Formula</b>                      | N2 Ni O6 . 6 H2 O        |
| <b>Molecular Weight</b>                       | 290.8                    |

## 10. Stability and reactivity

|                               |                                                                                            |
|-------------------------------|--------------------------------------------------------------------------------------------|
| <b>Reactive Hazard</b>        | Yes                                                                                        |
| <b>Stability</b>              | Oxidizer: Contact with combustible/organic material may cause fire.                        |
| <b>Conditions to Avoid</b>    | Avoid dust formation. Incompatible products. Excess heat. Combustible material.            |
| <b>Incompatible Materials</b> | Strong oxidizing agents, Organic materials, Finely powdered metals, Acids, Strong reducing |

agents, Combustible material

**Hazardous Decomposition Products** nitric acid**Hazardous Polymerization** No information available.**Hazardous Reactions** None under normal processing.

## 11. Toxicological information

### Acute Toxicity

#### Product Information Component Information

| Component                               | LD50 Oral                 | LD50 Dermal | LC50 Inhalation |
|-----------------------------------------|---------------------------|-------------|-----------------|
| Nickel(II) nitrate, hexahydrate (1:2:6) | LD50 = 1620 mg/kg ( Rat ) | Not listed  | Not listed      |

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Irritation** Severe eye irritant; Irritating to skin**Sensitization** May cause sensitization by skin contact**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component                               | CAS-No     | IARC       | NTP   | ACGIH      | OSHA       | Mexico     |
|-----------------------------------------|------------|------------|-------|------------|------------|------------|
| Nickel(II) nitrate, hexahydrate (1:2:6) | 13478-00-7 | Group 1    | Known | Not listed | X          | Not listed |
| Nickel(II) nitrate                      | 13138-45-9 | Not listed | Known | Not listed | Not listed | Not listed |

IARC (International Agency for Research on Cancer)

IARC (International Agency for Research on Cancer)

Group 1 - Carcinogenic to Humans

Group 2A - Probably Carcinogenic to Humans

Group 2B - Possibly Carcinogenic to Humans

**Mutagenic Effects** Possible risk of irreversible effects**Reproductive Effects** May cause harm to the unborn child.**Developmental Effects** No information available.**Teratogenicity** No information available.**STOT - single exposure** Respiratory system**STOT - repeated exposure** Lungs**Aspiration hazard** No information available**Symptoms / effects, both acute and delayed** Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing**Endocrine Disruptor Information** No information available**Other Adverse Effects** See actual entry in RTECS for complete information.

## 12. Ecological information

### Ecotoxicity

Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

**Persistence and Degradability** Soluble in water Persistence is unlikely based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Will likely be mobile in the environment due to its water solubility.

### 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

### 14. Transport information

#### DOT

UN-No UN2725  
 Proper Shipping Name NICKEL NITRATE  
 Hazard Class 5.1  
 Packing Group III

#### TDG

UN-No UN2725  
 Proper Shipping Name NICKEL NITRATE  
 Hazard Class 5.1  
 Packing Group III

#### IATA

UN-No UN2725  
 Proper Shipping Name NICKEL NITRATE  
 Hazard Class 5.1  
 Packing Group III

#### IMDG/IMO

UN-No UN2725  
 Proper Shipping Name NICKEL NITRATE  
 Hazard Class 5.1  
 Packing Group III

### 15. Regulatory information

#### International Inventories

| Component                               | CAS-No     | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|-----------------------------------------|------------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Nickel(II) nitrate, hexahydrate (1:2:6) | 13478-00-7 | -   | -    | -    | -                                             | -         | -      | -   |
| Nickel(II) nitrate                      | 13138-45-9 | X   | -    | X    | ACTIVE                                        | 236-068-5 | -      | -   |

| Component                               | CAS-No     | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|-----------------------------------------|------------|-------|----------|------|------|------|------|-------|-------|
| Nickel(II) nitrate, hexahydrate (1:2:6) | 13478-00-7 | X     | -        | -    | -    | X    | X    | X     | X     |
| Nickel(II) nitrate                      | 13138-45-9 | X     | KE-25844 | X    | X    | X    | X    | X     | X     |

#### Legend:

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

#### Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

| Component                               | Canada - National Pollutant Release Inventory (NPRI) | Canadian Environmental Protection Agency (CEPA) - List of Toxic Substances | Canada's Chemicals Management Plan (CEPA) |
|-----------------------------------------|------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------|
| Nickel(II) nitrate, hexahydrate (1:2:6) | Part 1, Group A Substance                            |                                                                            |                                           |
| Nickel(II) nitrate                      | Part 1, Group A Substance                            |                                                                            |                                           |

#### Other International Regulations

#### Authorisation/Restrictions according to EU REACH

| Component                               | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                                                                                                                                                         | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------------------------------------|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Nickel(II) nitrate, hexahydrate (1:2:6) | -                                                                   | Use restricted. See item 27. (see link for restriction details)                                                                                                                                                                                                       | -                                                                                                     |
| Nickel(II) nitrate                      | -                                                                   | Use restricted. See item 28. (see link for restriction details)<br>Use restricted. See item 30. (see link for restriction details)<br>Use restricted. See item 75. (see link for restriction details) Use restricted. See item 27. (see link for restriction details) | -                                                                                                     |

#### REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

#### Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component                               | CAS-No     | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-----------------------------------------|------------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Nickel(II) nitrate, hexahydrate (1:2:6) | 13478-00-7 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |
| Nickel(II) nitrate                      | 13138-45-9 | Listed         | Not applicable               | Not applicable            | Not applicable                             |

| Component                               | CAS-No     | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-----------------------------------------|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Nickel(II) nitrate, hexahydrate (1:2:6) | 13478-00-7 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Nickel(II) nitrate                      | 13138-45-9 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

#### Prepared By

Product Safety Department  
Email: chem.techinfo@thermofisher.com  
www.thermofisher.com

#### Creation Date

23-December-2009

#### Revision Date

29-March-2024

#### Print Date

29-March-2024

#### Revision Summary

New emergency telephone response service provider.



**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**